

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Wire armored and PVC Sheathed

Voltage Grade

18 / 30 (36) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum tape and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502 or BS 6622
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Aluminum Wires applied over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



MEDIUM VOLTAGE CABLES

Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
	mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km
A - Copper Conductors													
PM-CT14-X9-01-RP	50	0.3870	0.4937	0.4937	0.1436	0.5503	0.5041	228	220	244	241	37.7	1855
PM-CT15-X9-01-RP	70	0.2680	0.3420	0.3420	0.1615	0.5201	0.4738	275	268	303	301	39.5	2135
PM-CT16-X9-01-RP	95	0.1930	0.2465	0.2465	0.1762	0.4990	0.4528	323	318	366	365	41.2	2450
PM-CT17-X9-01-RP	120	0.1530	0.1956	0.1957	0.1898	0.4862	0.4399	360	358	419	418	44.0	2915
PM-CT18-X9-01-RP	150	0.1240	0.1588	0.1588	0.2042	0.4705	0.4243	394	398	473	468	45.5	3315
PM-CT19-X9-01-RP	185	0.0991	0.1272	0.1274	0.2215	0.4544	0.4082	443	445	539	530	47.3	3745
PM-CT20-X9-01-RP	240	0.0754	0.0973	0.0974	0.2434	0.4373	0.3911	492	503	628	623	50.0	4425
PM-CT21-X9-01-RP	300	0.0601	0.0781	0.0783	0.2643	0.4239	0.3777	545	556	712	689	52.4	5115
PM-CT22-X9-01-RP	400	0.0470	0.0619	0.0622	0.2898	0.4100	0.3638	592	614	808	768	55.3	6140
PM-CT23-X9-01-RP	500	0.0366	0.0493	0.0498	0.3209	0.3957	0.3494	646	675	914	851	58.8	7350
PM-CT24-X9-01-RP	630	0.0283	0.0394	0.0401	0.3594	0.3843	0.3381	700	735	975	995	63.1	8835
PM-CT25-X9-01-RP	800	0.0221	0.0323	0.0332	0.4006	0.3711	0.3249	743	791	1065	1106	67.7	10815
PM-CM26-X9-01-RP	1000	0.0176	0.0249	0.0251	0.4921	0.3672	0.3210	818	874	1216	1264	78.9	14175
B - Aluminum Conductors													
PM-AT14-X9-01-RP	50	0.6410	0.8219	0.8219	0.1476	0.5426	0.4964	180	171	191	188	38.1	1595
PM-AT15-X9-01-RP	70	0.4430	0.5682	0.5682	0.1624	0.5175	0.4713	218	209	236	234	39.6	1725
PM-AT16-X9-01-RP	95	0.3200	0.4105	0.4106	0.1781	0.4964	0.4502	258	249	285	284	41.4	1900
PM-AT17-X9-01-RP	120	0.2530	0.3247	0.3247	0.1907	0.4851	0.4388	289	282	331	326	44.1	2210
PM-AT18-X9-01-RP	150	0.2060	0.2645	0.2645	0.2071	0.4677	0.4215	320	313	374	370	45.8	2455
PM-AT19-X9-01-RP	185	0.1640	0.2108	0.2109	0.2244	0.4520	0.4058	357	353	428	424	47.6	2655
PM-AT20-X9-01-RP	240	0.1250	0.1610	0.1611	0.2434	0.4373	0.3911	402	404	503	500	50.0	2970
PM-AT21-X9-01-RP	300	0.1000	0.1291	0.1293	0.2643	0.4239	0.3777	442	450	574	561	52.4	3270
PM-AT22-X9-01-RP	400	0.0778	0.1009	0.1011	0.2898	0.4100	0.3638	489	505	661	639	55.3	3775
PM-AT23-X9-01-RP	500	0.0605	0.0792	0.0795	0.3209	0.3957	0.3494	539	565	760	722	58.8	4295
PM-AT24-X9-01-RP	630	0.0469	0.0622	0.0627	0.3604	0.3839	0.3377	605	625	833	840	63.2	4970
PM-AT25-X9-01-RP	800	0.0367	0.0499	0.0505	0.4006	0.3711	0.3249	663	691	942	956	67.7	5750
PM-AM26-X9-01-RP	1000	0.0291	0.0389	0.0390	0.4921	0.3672	0.3210	740	778	1087	1114	78.9	7545

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 19/33 (36) KV